Ritvik Sharma

CSE-DS

Company: IDFC First Bank

Job Profile: Application Engineer

Offer Type: Internship + Placement

Internship Stipend: Not Specified (Heard 40k)

CTC: 18.1 LPA (14L Base + 2L Signing bonus + 2.1L Variable Pay)

Location: Not Specified (Heard Internship in Bangalore and Job in Mumbai)

Process Date: 28th July - 30th July

Placement Process:

- The Placement process started with the announcement of the PPT (pre-placement talk) which was in the second last week of July. The dates of OA (online assessment) were announced shortly and were scheduled for the next week.
- A total of 4 interview rounds were conducted, excluding the online assessment. There were 2 technical rounds, 1 HR discussion round and the last Business round which turned out to be a technical and curriculum discussion round for me.
- The placement process for IDFC is too hectic because of so many rounds and the rigorous nature of interviews. I would advise not to go lightly and do not hope for any chill interview rounds, the selection is also too strict and depends entirely on the first 2 technical rounds. The people who passed the technical interviews and appeared for the HR round were all selected except a few people which was really disheartening.

1. **<u>Online Assessment</u>** [135 candidates appeared for this round]

    - This round was conducted on the campus, it was around 1hr 15 minutes test. The criteria for this round was on the basis of branches. EXTC was not allowed to sit for the process and were informed almost 10-15 minutes prior to the OA.
    - The test consisted of 3 sections, General aptitude, CS fundamentals, DSA
    - In the aptitude section, normal aptitude questions like probability, profit loss, worker problem, speed and time were asked. There were around 15 questions and we were given 25 minutes to solve them. The aptitude test I felt was easy and was solvable (I hate aptitude and do not practice for it and just tend to skip them in the test if I could not get the idea by seeing the question (absolutely not recommended ❌❌❌)). For aptitude refer indiabix website, and practice quantitative and pattern recognition questions.
    - The next section was on CS fundamentals. It included questions from DBMS, OS, CCN, OOPS. Basic questions were asked in this section, such that if you have a good knowledge of your syllabus from the 2nd year then you would be able to tackle this section fairly. You can refer to Babbar's youtube channel for DBMS, OS and Kunal Kushwaha for OOPS.

- The third section was DSA. The questions asked in this section were fairly straightforward and were solvable by any brute force idea. The problem with this section was with the different programming languages (ideally should not happen, but happened with Java and Python people). I did not have to think on this section as I directly implemented the solutions.
    1. The first problem was that we needed to find the sum of good numbers from a range [l, r], a good number is a number in which the difference between the sum of all the digits in the number and the largest digit is exactly equal to the largest digit. Eg 3003 ((3+0+0+3)-3 == 3). The brute force idea worked but gave TLE in most languages like C++, Java. I implemented it in C and was able to pass all the test cases.
    2. The second problem was to find out that after sorting an array how many elements did not change their position. Again a fairly simple problem and just using the inbuilt sort function worked, I did it in C++ and passed all the test cases. Again, it was a problem with Java and Python people with the input and test cases.

For passing the online assessment, I recommend practicing aptitude and do not skip it like I did because solving aptitude along with DSA will help you get an edge over others. Do leetcode, codeforces, codechef or any other platform that you like, DO NOT skip over any advanced DSA topic however difficult it may seem (cost me my phonepe interview had I known lifting beforehand). Solving advanced DSA problems helps you build the instinct where you do not have to put efforts into coming up with a solution.

After this round around 45 candidates were selected. The mistakes in the DSA section were taken into account by the company and the students who wrote a correct solution were selected.

2. <u>**Technical Round 1**</u> [45 candidates appeared for this round]

- This round was conducted on campus on 30th July in the morning. There were 9 panels and each panel had 2 interviewers. This round mainly focused on the core CS knowledge of the candidates like basic data structures, OOPS, Schema Design, etc. The round lasted around 45-50 minutes.
- When I entered the room, they shook hands with me and told me to sit. They did not ask for my introduction (I was not asked to introduce myself in any interviews I gave (don't know the reason behind that)). They took a copy of my resume, took a look at it and then asked whether I am ready to start with the interview.
- They started with very very basic questions on data structures like what are the types of data structures (linear, non-linear), when to use which, what is stacks and queues, how will you implement stack using queues and vice-versa, what are linked lists and their types. They were very basic questions and if you have even the lowest level of knowledge then you should be able to answer their questions without hesitation. (I would suggest when you are definitely clear with the answer, make some stories and tell, do not tell directly, this allows you to cover time and list down some extra points in front of the interviewer which many candidates would not have mentioned had they been in a hurry.)
- Then he asked me about my primary programming language. I said I use C++ for problem solving, Javascript and Nodejs for I/O intensive developmental tasks and Java Springboot for CPU heavy developmental tasks. So they went ahead to ask about OOPS of C++ and Java. I

was told to explain every concept with respect to both of the languages and highlight the differences between them.

- He asked me about polymorphism, inheritance, the diamond problem of multiple inheritance and how it is handled in Java and C++, friend functions and classes with their counterparts in Java. When to use Java, when to use C++. He basically asked me the whole of OOPS, in about 25 minutes and I was able to answer all of the questions.
- Next he gave me a situation of an online shopping platform which will sell products of all the types like electronics, clothes, food and everything, basically Amazon. He asked me how I would design the schema of this platform and handle the categories of products. He asked me how I will make the foreign key relations, what kind of attributes I store and how many tables would I require.
- Then on this schema he asked me to find products of a particular category, a particular brand, and above a particular rating. He told me to write a query on that and I was able to write the query (it was a simple query of inner join with some where clauses).
- Next he asked me how I track the trends of the customers who visited the platform. To this I answered by keeping the search history, keywords, order details and similar preferences of the user in a vectorized form inside the database and using it to predict the preferences of the users. I went a bit into machine learning for this question and was able to tell them the flow on how we will gather the data of user behaviour and use it to suggest similar products to the customer to boost the business.

For the technical round, I would suggest having a strong knowledge of the CS fundamentals and programming languages on your finger tips. It is not an exaggeration, this is what makes a difference while giving the feedback. Be calm in the interview and answer every question to the point, by making some stories to sugarcoat your answer. If you absolutely know the answer to a question, cook the response as much as you can, do not give straightforward answers like some AI or a robot. Keep your theory knowledge crisp with reasons for every fact, you need to know what is happening and why something is made in a certain way.

After this round around 22 candidates were shortlisted for the next round

3. **<u>Technical Round 2</u>** [22 candidates appeared for this round]

- This round was conducted on campus on 30th July. There were 6 panels and each panel had 2 interviewers. This round focused on core development knowledge, system design and problem solving. This round lasted around 45 minutes and was fast paced. We were expected to answer quickly like a rapid fire round, the interviewer did not show any kind of expression from which students could make out whether they were satisfied with the answer or not.
- They started by asking me about my day, I told them it was going fine uptill now. Then they proceeded to test my developmental knowledge by asking deep questions like what is actually nodejs and why it was developed. Then they proceeded to ask some system design questions like how would you scale a system if a lot of users came to your application. They did not even listen to the full answer, interrupted between every question and changed to the

next question abruptly. This would have been a tactic to test the candidate in difficult situations and how they would react under pressure.

- I heard from other candidates that they were taunting the students like "you fumbled here how would you handle in IDFC" or "you have mentioned so and so technology on your resume then you should be aware of it".
- They then proceeded to ask me a DSA question, only my panel was asking DSA. The problem required me to find the longest palindromic substring in a given string. The problem could be found on leetcode (5th problem on leetcode). They told me I have 15 minutes to code for the solution. The panelist handed me their macbook and told me to write the code on an online compiler. I had to write in codeforces style i.e. taking the input and then printing the longest palindromic substring. They did not care about the most optimal approach or the time and space complexity. They just wanted a working code and also did not provide any details about the question like constraints and all. After coding they asked me to explain the approach I took. It was a fairly simple problem and I was able to solve it in about 10 minutes. Thinking out loud, or explaining while coding was not at all required, their sole purpose was to get a working code.
- After this they tried to run the code but the online compiler was not working due to some internet issues. In spite of this they told me that my code is incorrect and has errors that is why it was not working. I argued with them that the code I wrote could not be incorrect and made them try running the code meanwhile the other interviewer went on to ask further questions.
- He asked me a puzzle. I have coins of denominations {1,2,5} and had to make a sum by taking at least one coin from every denomination. There was no logic involved in this puzzle and I just had to mod the sum by every denomination. While I was answering this the other interviewer finally ran the code by installing a GCC compiler on his macbook and accepted that the code was working perfectly. This gave me an edge as the other candidates in my panel faced a similar issue of the online compiler not working.

Be prepared to face tougher situations as the purpose of further rounds is to test how mentally strong you are. DO NOT MELT IN FRONT OF THE INTERVIEWER HOWEVER RUDE OR UNINTERESTED THEY MAY SEEM. If the interviewer interrupts you in between, take it as a positive point because you were already on the right track and they got it, that is why they change the questions. Have confidence in yourself and do not underestimate yourself, if you are sure of something then do argue with the interviewer, sometimes they are also at fault.

After this 9 candidates were selected for the HR round.

4. **<u>HR Round</u>** [9 candidates appeared for this round]

- The HR round was pretty chill. She was taking around 45 minute interviews of the candidates but due to time constraints (it was past 6 in the evening), she took the remaining candidates' interviews for about 15-20 minutes each.
- She started by asking how my day was, when I came to college for the interviews and how was the interview process in general. She asked where I stay and asked what my parents do.
- She asked me pretty straightforward behavioral and situational questions. She started by asking what if the manager is out of reach and does not reply, what will you do. Then she asked what if the manager micro-manages you, what will you do in this scenario.

- She also asked what would be my melting point (the point where you feel you will leave the company). All the questions were pretty standard, and I used pretty standard answers that chatGPT gives when we ask it to prepare for the HR round.
- The candidates who got shortlisted for the HR round were most probably guaranteed of the job, so the HR round was not very serious. The HR lady on the outside told us not to worry, they will try their best not to reject anyone after this.

Prepare for the HR round by chatGPT. They ask pretty standard questions and you are expected to give their standard answers as well, like a viva. Try to stay calm and smile in front of the HR

All 9 candidates appeared for the next round.

5. **Business Round** [9 candidates appeared for this round]

- This round was like a mere formality in the beginning as we have come a long way in clearing three intense interviews throughout the day. The interview was about 30 minutes and this was conducted in online mode because it was too late in the evening. Initially it was planned to take this round for half of the candidates on the next day, but we insisted to the HR lady (she was very friendly) to take it and complete it today itself, and they did so.
- Initially we were completely unaware as to what will be asked in this round as the HR also did not have any idea about the business round. She told us it may be related to the college curriculum that we have studied till now or something like that.
- The round for me turned out to be another technical round. The interviewer first asked me about my favorite subjects in college to which I answered DBMS, OS, Distributed Computing. So he began asking about DC. He asked me about system scalability, design patterns and concurrency in the case of distributed systems. I was well versed with DC in my sem 6 so I was able to answer all the questions pretty easily.
- He then asked me the difference between frontend and backend (I don't know why such a question), so based on my answer he started cross questioning me about the difference between UI/UX and some other jargon. I was able to answer all the questions but felt that the interview didn't go as intended.

Be ready for surprises as this round was not initially planned. IDFC is known to have many interview rounds before selecting the candidates. Also do not assume that if you have come this far then it is absolutely guaranteed that you will get selected, as one candidate was not selected after these 4 rigorous interview rounds. I don't know the reason for that but it was pretty unfortunate.

Finally 8 candidates were finally selected for IDFC.

Useful Tips:

- Start early with DSA and development. It gives you a very strong edge over others. You may think that you have a lot of time, indeed it is true, but do not underestimate the power of consistency and long time efforts. Most people do not start preparation until they have completed their 5th semester. I started in the 3rd semester with DSA and development in the 4th semester.

- Do not be stuck in the "Dev or DSA" hell. YOU MUST HAVE BOTH OF THEM AT A PRETTY GOOD LEVEL. Nowadays everybody knows the MERN stack and it has become the new normal, do not limit yourself to only one technology. Explore other technologies as well like Java Development, Spring boot, etc. Make some projects which actually solve real world problems, not just a CRUD application with some eye-catching UI which is made by vercel v0 or claude.
- Participate in hackathons and solve their problem statements, even if you cannot win the hackathon do not stop building the application for the problem. You will anyways have a solid project with you if you keep on working with the statement.
- Do not ignore system design. It is now becoming common to ask system design to freshers and may happen that it will become a part of your process like DSA. At least know the basics of HLD, LLD and have a knowledge about the famous questions of system design. Use Aryan Mittal for system design.
- Give leetcode/codeforces/codechef/atcoder contests, they help you keep your mind in the correct momentum which is needed during the placement season. You cannot stop your mind midway through the season. Having internships with proper firms gives you an edge, because nowadays everybody has some kind of internship experience. If you do not have an internship, then you have to work a little more to compete with the ones having internships. Do not get inferiority complex with the people who have summer internships or any other. Hard work and consistency makes a lot of difference.
- DO NOT EXPECT THE PLACEMENT SEASON TO BE VERY SMOOTH. You have to be mentally strong for this and rejections are a part of it. I messed up my PhonePe interview because I did not know a specific advanced DSA concept (Binary Lifting) which led to my rejection. Do not think that advanced topics are not required for freshers. You never know what is waiting for you.
- Do not follow just one DSA sheet. One sheet may have concepts which the other may not. Go with striver, babbar for DSA. For CS fundamentals use Babbar, Kunal Kushwaha for OOPS. Do competitive programming, use TLE eliminators for that. Use TLE post contest discussion, they usually have solutions of up to 4-5 problems. Even if you have a low rating, watch and solve the later problems as they involve some good concepts which can sometimes pop up in front of you, and your mind gets trained to handle tougher problems.
- Keep your CGPA good enough, at least try to keep it above 8, it will help you to sit for all the companies. Sometimes the outcome is not in your hands, you may not pass the OA even after completely solving all the questions (happened to me with Zeta). Just make peace with it and look for the next company. Don't give a second thought to what is in the past. Don't let it bother you.
- Keep working and you will make it where you deserve. Hard work can never go in vain. Do not fall into that hardwork vs smartwork and all that superficial stuff. Keep it simple and drag yourself through it. Do not jumble yourself in tutorial hell, how to give interviews, how to smartly approach a problem, bla bla bla. Just be you and practice hard.

You may contact me here: Linkedin   Resume: Link

All the best.